

is modular design? It is an approach that has proven itself indispensable in the world of engineering long before the first computers were even built. Modular design means that an exceptionally complex system is broken down into manageable smaller parts or components. These smaller parts are called modules. These components can be created independently and can even be tested the same way. In a number of cases, given their lack of dependency on other modules, they can be used in more than one system.

Come to think of it, there is hardly any product out there today that does not rely on the concept of modularisation – cars, bikes, mobile phones, etc. Computers are modularised to the core. So when the hardware uses this concept so heavily, it becomes an absolute necessity for the software that is running on this hardware to also be modularised.

If your ultimate goal is to develop programs that are readable, reliable and even easy to maintain without too much effort from any end, you will need to employ some sort of modular software design. This becomes even more important if you are working on an application that is rather large in size. Modular programming will allow you to split your code into separate parts. These tinier parts are then called modules. The basic goal is to ensure that the separation of modules is real - there should be no or very negligible amount of dependencies upon other modules. When it comes to creating a stellar modular system, a number of modules will be created and built separately as well as independent of each other. The final application will be created by fixing all these modules together. There are a number of advantages to modularising code when working on a large application -

Keeps Things Simple

A module will typically focus on one relatively small portion of a problem as opposed to focusing on the entire problem at hand. If you are working on a single module, it gives you the choice to have a smaller problem domain that you need to wrap your head around. This ends up simplifying the entire process of development and is more likely to leave you with a process that is less prone to error or misjudgements.



Everything from mobile phones to cars are modularised these days, so why should your software be far behind?